KAIZEN MATHS

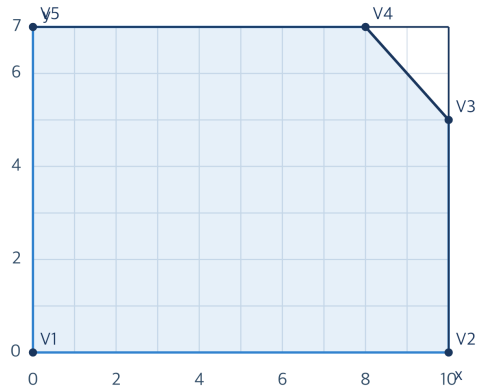
Lesson starter

Linear Programming

Optimisation

Name _____

SA1



Maximise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 7$

$x + y \leq 15$

Maximise P = 8x + 8y

ALG-AFEA-L2 · SA

Show your working

KAIZEN MATHS

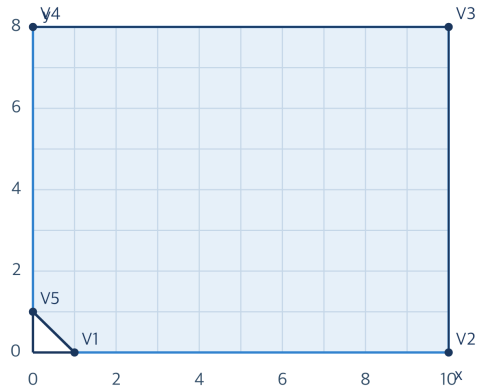
Lesson starter

Linear Programming

Optimisation

Name _____

SB1



Minimise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 8$

$x + y \geq 1$

Minimise P = 7x + 2y

ALG-AFEA-L2 · SB

Show your working

KAIZEN MATHS

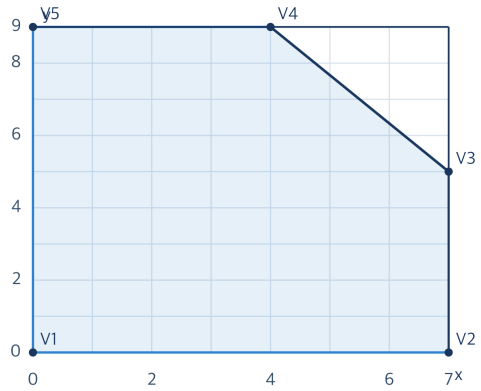
Lesson starter

Linear Programming

Optimisation

Name _____

SC1



Calculate P at each vertex and state which vertex gives the largest value.

$0 \leq x \leq 7$

$0 \leq y \leq 9$

$4x + 3y \leq 43$

$P = 4x + 2y$

ALG-AFEA-L2 · SC

Show your working

KAIZEN MATHS

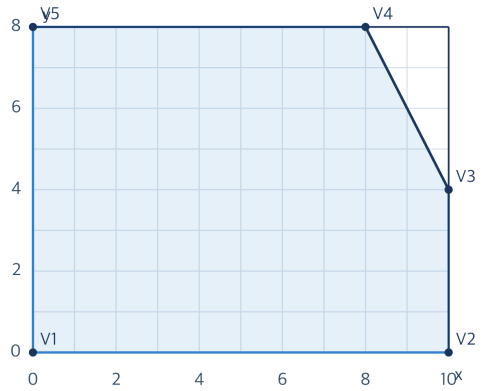
Lesson starter

Linear Programming

Optimisation

Name _____

SD1



Maximise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 8$

$2x + y \leq 24$

Maximise P = 2x + 6y

ALG-AFEA-L2 · SD

Show your working

KAIZEN MATHS

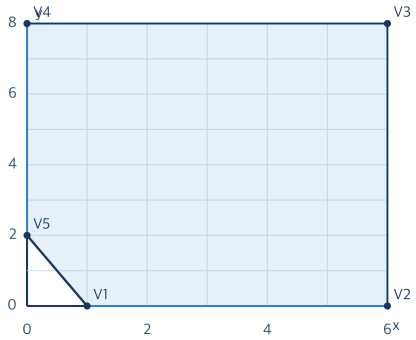
Exit ticket

Linear Programming

Optimisation

Name _____

EA1



Minimise the objective function over the feasible region.

$0 \leq x \leq 6$

$0 \leq y \leq 8$

$2x + y \geq 2$

Minimise P = 2x + 3y

ALG-AFEA-L2 · EA

Show your working

KAIZEN MATHS

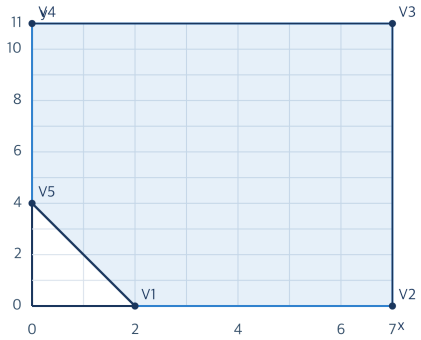
Exit ticket

Linear Programming

Optimisation

Name _____

EB1



Calculate P at each vertex and state which vertex gives the largest value.

$0 \leq x \leq 7$

$0 \leq y \leq 11$

$4x + 2y \geq 8$

$P = 9x + 2y$

ALG-AFEA-L2 · EB

Show your working

KAIZEN MATHS

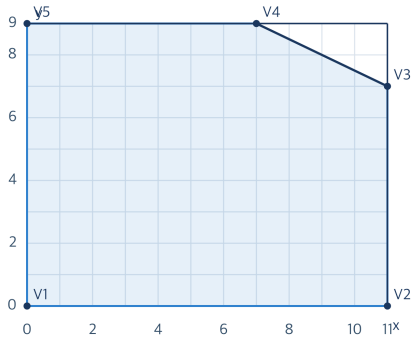
Exit ticket

Linear Programming

Optimisation

Name _____

EC1



Maximise the objective function over the feasible region.

$0 \leq x \leq 11$

$0 \leq y \leq 9$

$2x + 4y \leq 50$

Maximise P = 7x + 4y

ALG-AFEA-L2 · EC

Show your working

KAIZEN MATHS

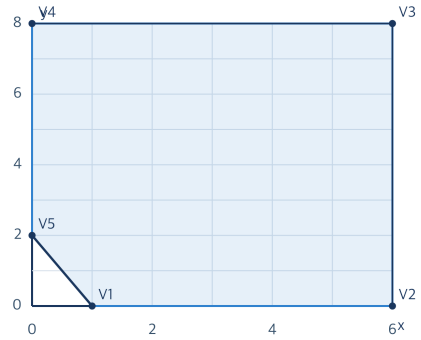
Exit ticket

Linear Programming

Optimisation

Name _____

EA1



Minimise the objective function over the feasible region.

$0 \leq x \leq 6$

$0 \leq y \leq 8$

$2x + y \geq 2$

Minimise P = 2x + 3y

ALG-AFEA-L2 · EA

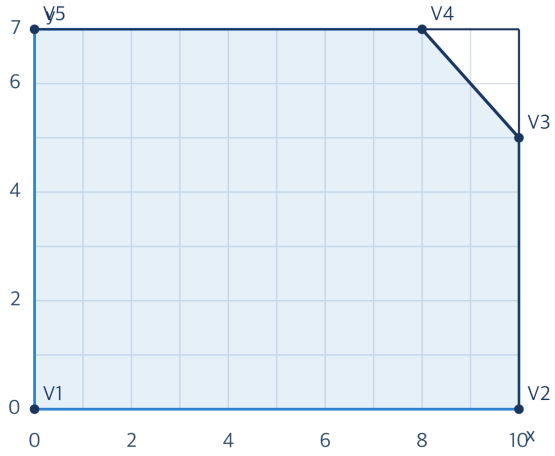
Show your working

Teacher answers & working

Linear Programming · Optimisation

ALG-AFEA-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Maximise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 7$

$x + y \leq 15$

Maximise P = 8x + 8y

Answer:

Maximum P = 120 at (10, 5)

Working:

A linear objective function reaches its optimum at a vertex of the feasible region.

Vertices: (0, 0), (10, 0), (10, 5), (8, 7), (0, 7).

Vertex	P	Value
(0, 0)	8(0) + 8(0)	0
(10, 0)	8(10) + 8(0)	80
(10, 5)	8(10) + 8(5)	120
(8, 7)	8(8) + 8(7)	120
(0, 7)	8(0) + 8(7)	56

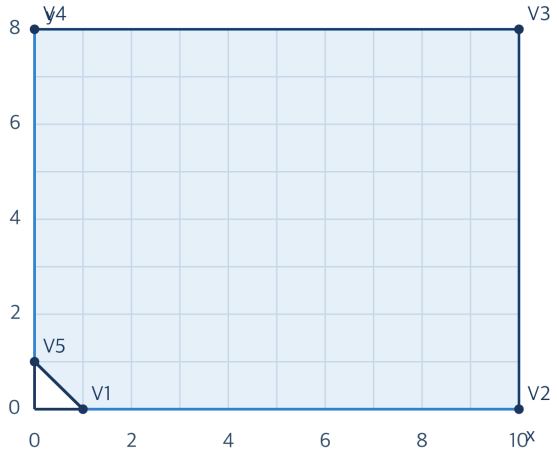
Maximum value is P = 120 at (10, 5).

Teacher answers & working

Linear Programming · Optimisation

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SB1



Minimise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 8$

$x + y \geq 1$

Minimise P = 7x + 2y

Answer:

Minimum P = 2 at (0, 1)

Working:

A linear objective function reaches its optimum at a vertex of the feasible region.

Vertices: (1, 0), (10, 0), (10, 8), (0, 8), (0, 1).

Vertex	P	Value
(1, 0)	7(1) + 2(0)	7
(10, 0)	7(10) + 2(0)	70
(10, 8)	7(10) + 2(8)	86
(0, 8)	7(0) + 2(8)	16
(0, 1)	7(0) + 2(1)	2

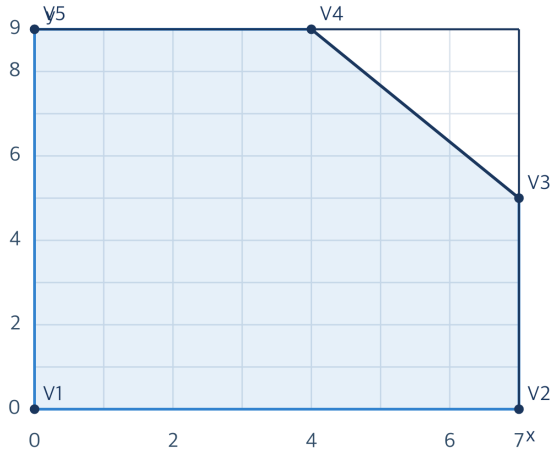
Minimum value is P = 2 at (0, 1).

Teacher answers & working

Linear Programming · Optimisation

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SC1



Calculate P at each vertex and state which vertex gives the largest value.

$0 \leq x \leq 7$

$0 \leq y \leq 9$

$4x + 3y \leq 43$

$P = 4x + 2y$

Answer:

Largest P = 38 at (7, 5)

Working:

Substitute each vertex into the objective function.

Vertex	P	Value
(0, 0)	$4(0) + 2(0)$	0
(7, 0)	$4(7) + 2(0)$	28
(7, 5)	$4(7) + 2(5)$	38
(4, 9)	$4(4) + 2(9)$	34
(0, 9)	$4(0) + 2(9)$	18

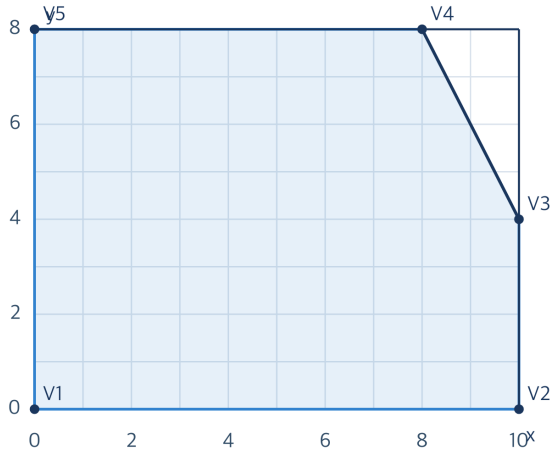
The largest value is P = 38 at (7, 5).

Teacher answers & working

Linear Programming · Optimisation

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SD1



Maximise the objective function over the feasible region.

$0 \leq x \leq 10$

$0 \leq y \leq 8$

$2x + y \leq 24$

Maximise P = 2x + 6y

Answer:

Maximum P = 64 at (8, 8)

Working:

A linear objective function reaches its optimum at a vertex of the feasible region.

Vertices: (0, 0), (10, 0), (10, 4), (8, 8), (0, 8).

Vertex	P	Value
(0, 0)	2(0) + 6(0)	0
(10, 0)	2(10) + 6(0)	20
(10, 4)	2(10) + 6(4)	44
(8, 8)	2(8) + 6(8)	64
(0, 8)	2(0) + 6(8)	48

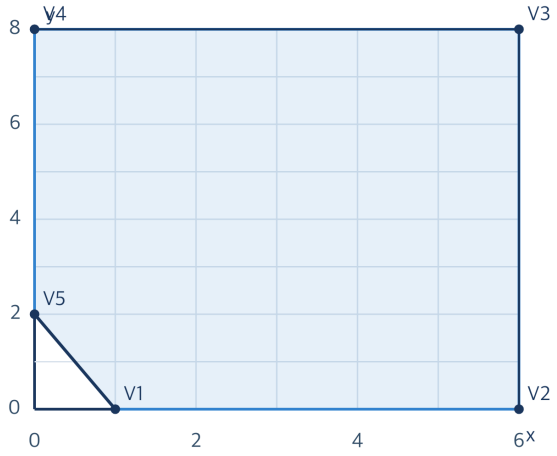
Maximum value is P = 64 at (8, 8).

Teacher answers & working

Linear Programming · Optimisation

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EA1



Minimise the objective function over the feasible region.

$0 \leq x \leq 6$

$0 \leq y \leq 8$

$2x + y \geq 2$

Minimise P = 2x + 3y

Answer:

Minimum P = 2 at (1, 0)

Working:

A linear objective function reaches its optimum at a vertex of the feasible region.

Vertices: (1, 0), (6, 0), (6, 8), (0, 8), (0, 2).

Vertex	P	Value
(1, 0)	2(1) + 3(0)	2
(6, 0)	2(6) + 3(0)	12
(6, 8)	2(6) + 3(8)	36
(0, 8)	2(0) + 3(8)	24
(0, 2)	2(0) + 3(2)	6

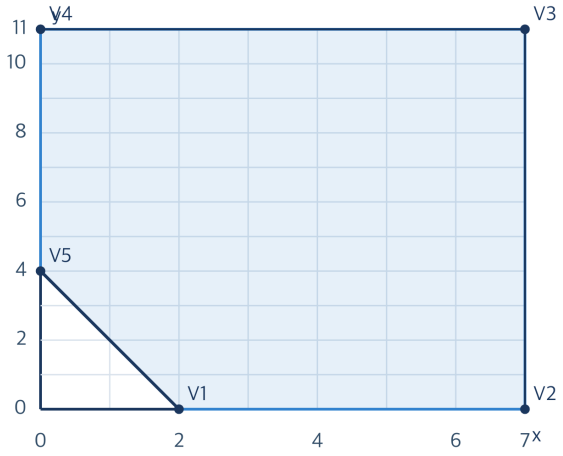
Minimum value is P = 2 at (1, 0).

Teacher answers & working

Linear Programming · Optimisation

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EB1



Calculate P at each vertex and state which vertex gives the largest value.

$0 \leq x \leq 7$

$0 \leq y \leq 11$

$4x + 2y \geq 8$

$P = 9x + 2y$

Answer:

Largest P = 85 at (7, 11)

Working:

Substitute each vertex into the objective function.

Vertex	P	Value
(2, 0)	$9(2) + 2(0)$	18
(7, 0)	$9(7) + 2(0)$	63
(7, 11)	$9(7) + 2(11)$	85
(0, 11)	$9(0) + 2(11)$	22
(0, 4)	$9(0) + 2(4)$	8

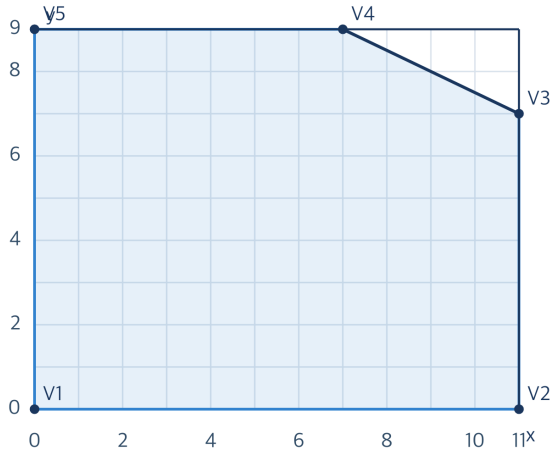
The largest value is P = 85 at (7, 11).

Teacher answers & working

Linear Programming · Optimisation

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EC1



Maximise the objective function over the feasible region.

$0 \leq x \leq 11$

$0 \leq y \leq 9$

$2x + 4y \leq 50$

Maximise P = 7x + 4y

Answer:

Maximum P = 105 at (11, 7)

Working:

A linear objective function reaches its optimum at a vertex of the feasible region.

Vertices: (0, 0), (11, 0), (11, 7), (7, 9), (0, 9).

Vertex	P	Value
(0, 0)	7(0) + 4(0)	0
(11, 0)	7(11) + 4(0)	77
(11, 7)	7(11) + 4(7)	105
(7, 9)	7(7) + 4(9)	85
(0, 9)	7(0) + 4(9)	36

Maximum value is P = 105 at (11, 7).